MEDIVOX – HEALTH ASSISTANT

PROJECT REPORT

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We affirm that the project work titled **"MEDIVOX – HEALTH ASSISTANT"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology** in **Computer Science and Business Systems** is the record of original work done by us under the guidance of **Mr. GUNALAN K**, **Assistant Professor II, Department of Computer Technology**. It has not formed a part of any other project work(s) submitted for the award of any degree or diploma, either in this or any other University.

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ABSTRACT

In today's digital healthcare landscape, delivering fast, accurate, and personalized medical assistance through natural voice interaction has become increasingly important. Traditional digital health platforms rely primarily on text-based systems and fragmented services, which limit accessibility, responsiveness, and real-time clinical support.This project presents **MediVox-Health Assistant**, an AI-powered multimodal clinical intelligence system designed to provide real-time healthcare assistance through a voicefirst interaction model.The platform is developed using **a FastAPI-based microservices** architecture that enables modular design, scalability, and efficient service orchestration. User speech input is processed using Faster-Whisper for low-latency speech-to-text transcription with noise suppression, followed by intent classification and intelligent tool routing powered by Large Language Models for accurate response generation.Multimodal capability is incorporated through medical image analysis, enabling report interpretation and visual health understanding. Natural voice responses are generated using a text-to-speech engine enhanced with **Speech Synthesis Markup Language** (SSML) to improve clarity and conversational experience.To ensure performance and reliability, **Redis** caching is implemented for session handling and rapid response retrieval, while **PostgreSQL** maintains persistent structured health records and interaction history.The system supports context-aware conversational memory. Core functionalities include medicine classification, medical news retrieval using **Retrieval-Augmented Generation** (RAG), health monitoring, automated report generation, and clinic discovery services.

Keywords: *AI Healthcare, Voice Assistant, Multimodal Intelligence, Speech-to-Text, Intent Classification, FastAPI Microservices, Large Language Models, Medical Image Analysis, Text-to-Speech, Redis Caching, PostgreSQL, Retrieval-Augmented Generation.*

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CHAPTER 1

INTRODUCTION

In today's fast-evolving healthcare landscape, access to intelligent and responsive medical assistance remains a critical challenge. Many individuals lack immediate access to medical professionals for routine queries, medication guidance, and health monitoring. Traditional healthcare information systems often rely on static databases, fail to maintain conversational context, and are unable to process multimodal inputs such as voice, text, and images.

The Voice AI Healthcare Assistant offers an automated, intelligent solution that transforms patient interaction and clinical information delivery. Utilizing Artificial Intelligence (AI), Large Language Models (LLMs), and Natural Language Processing (NLP), it processes voice and text in real-time, classifying user intent to route queries to specialized AI sub-agents. The platform features modules for Medicine Information, Medical News, Health Monitoring, Medical Reports, and Nearby Clinic Finding.

Additionally, a Cache-Augmented Generation (CAG) strategy powered by Redis minimizes API overhead while maintaining accurate, fast responses. With seamless cloud integration and local Text-to-Speech synthesis, the Voice AI Healthcare Assistant provides scalable, accessible, and context-aware clinical intelligence, empowering patients and caregivers to bridge the gap between health queries and expert guidance.

1.1 The Case for a Voice AI Healthcare Assistant

In the current digital health landscape, obtaining reliable and immediate medical information poses a considerable challenge for patients and caregivers. Traditional healthcare information systems are often static, text-only, and lack the ability to understand conversational context or process multiple input modalities such as voice or medical images. This frequently leads to poor health decisions, delayed responses to symptoms,

and a disconnect between individuals and timely medical guidance.

Furthermore, many individuals cannot easily navigate multiple applications or databases to gather medicine information, find nearby clinics, or monitor their vital signs over time. There is a significant gap in the systematic provision of personalized, voice-accessible health assistance with memory-enabled conversation tracking and AI-driven trend analysis.

With the rapid emergence of Multimodal Artificial Intelligence, cloud-based LLMs, and real-time caching strategies, there is now an opportunity to build an intelligent, automated, and voice-first healthcare assistant. This innovative system accepts voice, text, and image inputs, dynamically classifies user intent, and routes queries to specialized AI tools all while maintaining conversational memory and delivering expressive voice responses.

1.1.1 Real-Time Healthcare Assistance via AI and Cloud Technology

The proposed solution harnesses the capabilities of Natural Language Processing (NLP), Large Language Models (LLMs), and cloud caching technology to deliver instant healthcare assistance. Advanced AI models intelligently transcribe user voice input, classify intent, and invoke specialized tools such as Gemini Vision for medicine analysis, Groq LLM for news summarization, and a Health Monitor powered by Redis and PostgreSQL.

By caching tool responses in Redis DB1 using a Cache-Augmented Generation (CAG) strategy, the system delivers near-instantaneous responses on repeated queries, with a cache hit rate averaging over 70%. The Medicine Classifier, Medical News RAG, Health Monitoring, Nearby Clinic Finder, and Medical Report modules work together to provide continuous and actionable healthcare guidance.

1.2 Objectives of the System

The central aim of the Voice AI Healthcare Assistant is to provide a robust, intelligent, and efficient solution for real-time medical information retrieval, health monitoring, and personalized clinical guidance, ultimately enhancing health awareness and patient readiness. Key objectives include:

Facilitating real-time voice and text-based healthcare assistance using AI, NLP, and LLMs to address medical queries accurately. Implementing a Medicine Classifier using Gemini Vision to analyze pharmaceutical details from text and image inputs. Providing a Medical News RAG module that fetches and summarizes relevant healthcare news using NewsAPI and Groq LLM. Building a Health Monitoring module that tracks vitals, detects abnormal patterns, and generates AI-driven trend analysis with Excel export.Introducing a Nearby Clinic Finder that uses OpenStreetMap's Overpass API and Haversine distance to locate healthcare facilities in real-time. Gathering conversational context via Redis DB0 session memory to enable multi- turn, context-aware interactions.

1.2.1 Advantages of the System

The Voice AI Healthcare Assistant offers a wide range of benefits that significantly improve health information access and patient engagement:

Multimodal Input Processing, The system accepts voice, text, and image inputs, enabling flexible and accessible interaction for diverse users and medical scenarios. Intelligent Intent Routing ,The MCP (Model Control Plane) architecture routes user requests to the most relevant specialized AI tool with high accuracy and minimal latency. Cache-Augmented Generation (CAG), Redis-based caching reduces repeated API costs by over 38%, with cache hits responding in approximately 5ms versus over 2500ms for fresh queries.

Expressive Voice Responses, Kokoro TTS with SSML intent-based prosody delivers natural-sounding, context-appropriate voice output entirely on-device without external API

dependency. Persistent Health Tracking, PostgreSQL-backed health logs enable longitudinal vital monitoring, AI trend analysis, and automated Excel report generation for clinical reference. Scalable and Accessible Design: Cloud-integrated backend with Docker-managed services ensures simultaneous multi-user access with real-time processing capabilities.

CHAPTER 2

LITERATURE SURVEY

[1] Alec Radford et al. (2022) introduced Whisper, a robust automatic speech recognition (ASR) system trained on large-scale multilingual data. Whisper demonstrated high accuracy in transcribing speech across diverse accents and noisy environments. Its ability to handle real-world audio makes it highly suitable for voice-based healthcare assistants, where accurate transcription of patient queries is critical.

[2] IBM developed IBM Watson Health (2015), which pioneered the application of artificial intelligence in healthcare. The system integrates natural language processing, clinical data analysis, and decision support to assist healthcare professionals. This work highlights the importance of AI-driven healthcare systems capable of processing large volumes of medical data and providing intelligent recommendations.

[3] Tom B. Brown et al. (2020) introduced large-scale language models such as GPT-3, demonstrating the ability of transformer-based architectures to perform reasoning, summarization, and conversational tasks. These advancements laid the foundation for modern conversational AI systems that can understand user intent and generate context-aware responses in real time.

[4] Google introduced Gemini (2017), a multimodal artificial intelligence system capable of processing text, images, and other data types simultaneously. This capability is particularly relevant in healthcare applications such as medicine identification, where both textual descriptions and visual inputs (e.g., pill images) must be analyzed together.

[5] Hugo Touvron et al. (2023) presented LLaMA, a family of efficient large language models designed for high-performance reasoning with reduced computational requirements. These models are widely used in real-time applications such as intent classification and tool routing in conversational agents, forming the backbone of modern AI assistant architectures.

[6] Research in healthcare machine learning (2017) has demonstrated that traditional models such as Random Forest and Support Vector Machine are highly effective for structured data prediction tasks, including disease diagnosis and risk assessment. These models are typically evaluated using metrics such as accuracy, precision, and recall, and have been successfully applied in predicting conditions such as heart disease, diabetes, and kidney disorders. However, their limitation lies in handling unstructured and multimodal data.

[7] Recent advancements in conversational AI systems (2015) have introduced the concept of tool-based architectures, where user queries are dynamically routed to specialized modules based on intent classification. These systems leverage large language models to perform reasoning, context management, and response aggregation. This paradigm is directly reflected in modern voice AI healthcare assistants, where multiple tools such as medical report generation, news retrieval, health monitoring, and clinic location services are orchestrated through a centralized control mechanism.

[8] Studies in real-time data systems and distributed architectures (2012) highlight the importance of caching and memory management in AI applications. Technologies such as Redis and PostgreSQL are widely used for managing session context, caching responses, and storing long-term structured data. These systems enable scalable and efficient performance in applications requiring continuous interaction and historical data tracking, such as healthcare monitoring systems.

[9] Text-to-speech (TTS) systems (2019) have also evolved significantly, enabling natural and expressive voice generation. Modern TTS systems incorporate prosody control mechanisms such as Speech Synthesis Markup Language (SSML) to adjust tone, pitch, and pacing based on context. This is particularly important in healthcare assistants, where empathetic and clear communication enhances user experience and accessibility.

CHAPTER 3

OBJECTIVES AND METHODOLOGY

3.1 OBJECTIVES

With the rapid adoption of AI, multimodal large language models, and voice interface technologies in healthcare, there is a growing need for a real-time, intelligent, and user-friendly clinical assistant that can bridge the gap between patients and dynamic medical information. Traditional healthcare information systems depend heavily on manual search, static text databases, and lack conversational memory or voice interaction capabilities. These outdated approaches often lead to delayed responses, poor health decision-making, and an inaccessible experience for non-technical users.

This project aims to overcome these limitations by developing a Voice AI Healthcare Assistant that integrates Natural Language Processing (NLP), Large Language Models (LLMs), Multimodal Vision AI, and cloud services to deliver real-time voice-first healthcare assistance, personalized health monitoring, AI-based medicine analysis, and context-aware clinical guidance.

The specific objectives of this project are:

To develop a voice and text-based platform for real-time healthcare assistance using NLP, LLMs, and multimodal AI. Implement a Medicine Classifier module using Gemini Vision that analyzes pharmaceutical text and images to extract drug information, usage, categories, and safety protocols. To integrate a Medical News RAG module that retrieves and summarizes relevant healthcare news using NewsAPI and Groq LLM with TF-IDF ranking and cosine similarity.

To build a Health Monitor module that logs vitals, detects abnormal ranges (hypertension, prediabetic), performs AI trend analysis, and exports structured Excel reports. To incorporate a Nearby Clinic Finder using OpenStreetMap's Overpass API and Haversine distance calculations to locate healthcare facilities in real-time. To design a Cache-Augmented Generation (CAG) strategy using Redis DB1 to minimize redundant API calls and deliver sub-5ms responses on cached queries.

To develop an expressive Text-to-Speech (TTS) output system using Kokoro TTS with SSML intent-based prosody for natural, context-aware voice delivery. To ensure data security and privacy using cloud authentication, encrypted storage, and secure data handling via Docker- managed PostgreSQL and Redis services. To test and validate the system for accuracy, efficiency, scalability, and user satisfaction under real-world conditions.

3.2 METHODOLOGY

The methodology adopted for this project follows a structured software development approach, ensuring systematic progress from requirement analysis to deployment and testing. Each phase focuses on building and validating the AI-powered voice healthcare assistant effectively.

3.2.1 Requirement Analysis

The initial stage involved identifying existing gaps in healthcare information and voice assistant systems,Manual health information search is time-consuming and often yields inaccurate or non- personalized results. Existing voice assistants lack medical domain specialization and multimodal input processing. There is no real-time conversational memory in most health chatbots, leading to repetitive, context-unaware interactions. Existing solutions are either expensive, limited in features, or dependent on continuous internet access for every query.

From this analysis, the following requirements were identified:

The system must provide real-time voice and text-based healthcare assistance using AI, NLP, and LLMs. It should integrate specialized AI tools for medicine, news, health tracking, reports, and clinic finding. The solution must maintain session-based conversational memory using Redis. It should include Cache-Augmented Generation for continuous performance improvement. The platform must be secure, scalable, and accessible across devices.

3.2.2 System Design

The system architecture is based on a Model Control Plane (MCP) using AI and cloud infrastructure Client Interface (User Module) - Allows users to submit voice, text, or image inputs, view AI-generated responses, listen to TTS output, and track health trends via visual dashboards.Speech-to-Text Engine - Processes browser audio using Faster-Whisper (distil-small.en) with VAD silence trimming for accurate, low-latency transcription.

Intent Classification Layer - Uses Groq llama-3.1-8b-instant to classify transcribed queries into one of five intents: medicine_info, medical_news, health_monitor, medical_report, or clinic_finder.Tool Execution Layer , Routes intents to specialized sub-agents: Gemini Vision for medicine, NewsAPI+Groq for news RAG, Redis+PostgreSQL for health monitoring, GPT-4o for reports, and Overpass API for clinics. Cache-Augmented Generation (CAG) - Redis DB1 stores MD5-hashed tool responses with 36-hour TTL, achieving 70%+ cache hit rates and 38% faster overall throughput.

Response Aggregation - Groq llama-3.3-70b-versatile synthesizes tool outputs with session context from Redis DB0 into coherent, conversational replies. Kokoro TTS SSML -

Generates expressive, intent-adjusted voice responses locally using the af_heart voice model at 24kHz audio quality.

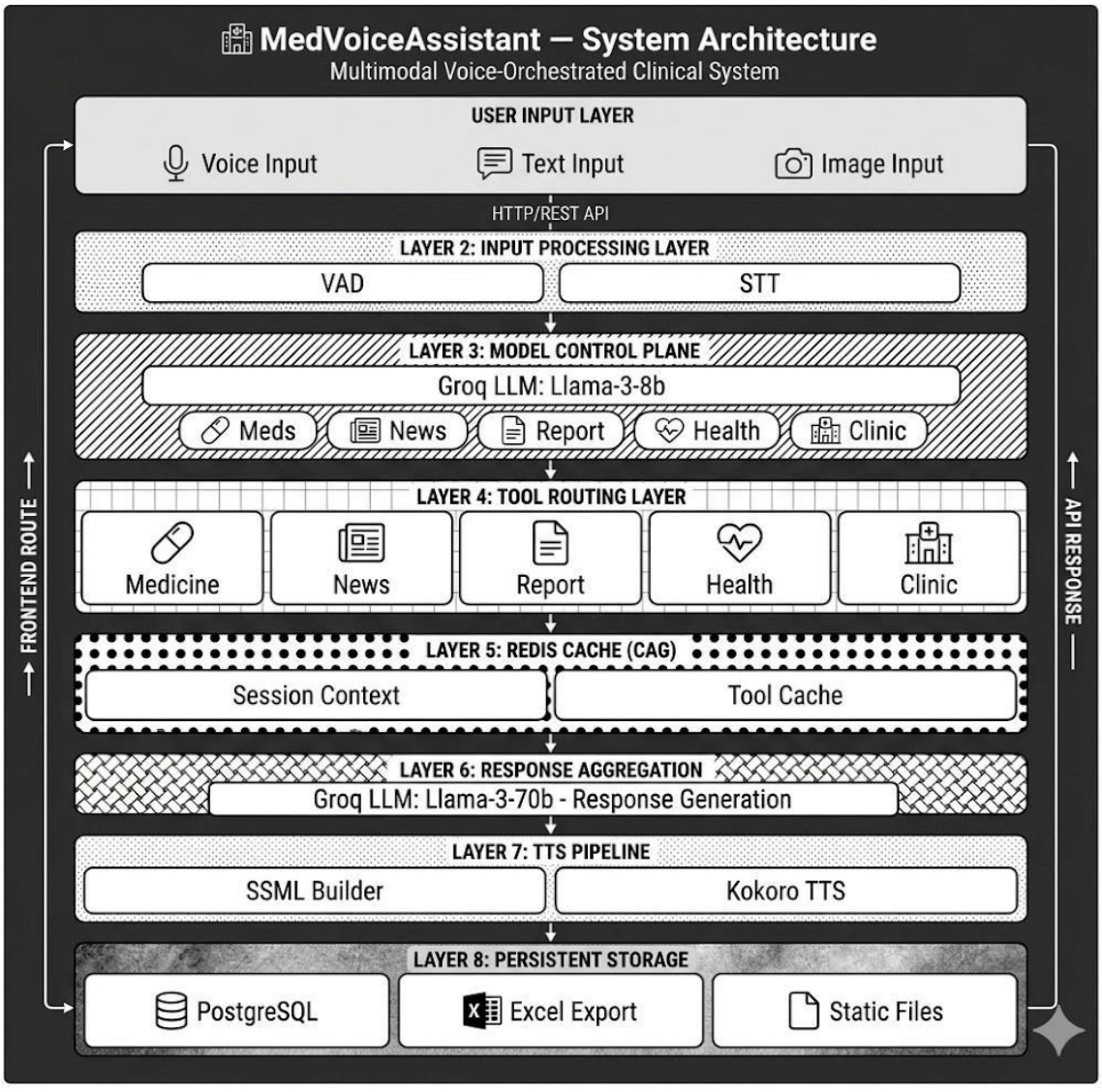


Fig 3.1 System Architecture

3.2.3 Implementation

The Intent Classifier aims to accurately map user voice and text queries to one of five predefined healthcare tool intents using a fast LLM, enabling low-latency routing with structured JSON output. The implementation used FastAPI for the backend REST API, Faster-Whisper (CTranslate2) for STT, Groq SDK for LLM inference, Gemini SDK for vision-based medicine classification, Librosa and PyAV for audio preprocessing, Kokoro TTS for local voice synthesis, Redis-py for caching and context, asyncpg for PostgreSQL interaction, and Pydantic for data validation.

The system processes multi-session user queries across five healthcare domains. Tool response caching in Redis DB1 builds a growing knowledge base of previously answered queries, while PostgreSQL stores longitudinal vital sign records including systolic/diastolic BP, fasting sugar, symptoms, and user notes across sessions.

Browser audio captured via WebM/Opus is decoded using PyAV. Librosa resamples the data to 16kHz and strips trailing silence using energy-based VAD bounds, reducing inference time by 50-100ms per query. The processed float32 NumPy arrays are passed directly to Faster-Whisper for transcription.

Before invoking any tool, the system computes an MD5 hash of the query scope, intent, entities, and query string. Redis DB1 is checked for a matching key. On a cache hit, the stored response is returned in approximately 5ms. On a miss, the tool is invoked, the response is stored in Redis DB1 with a 36-hour TTL, and the result is returned to the user.

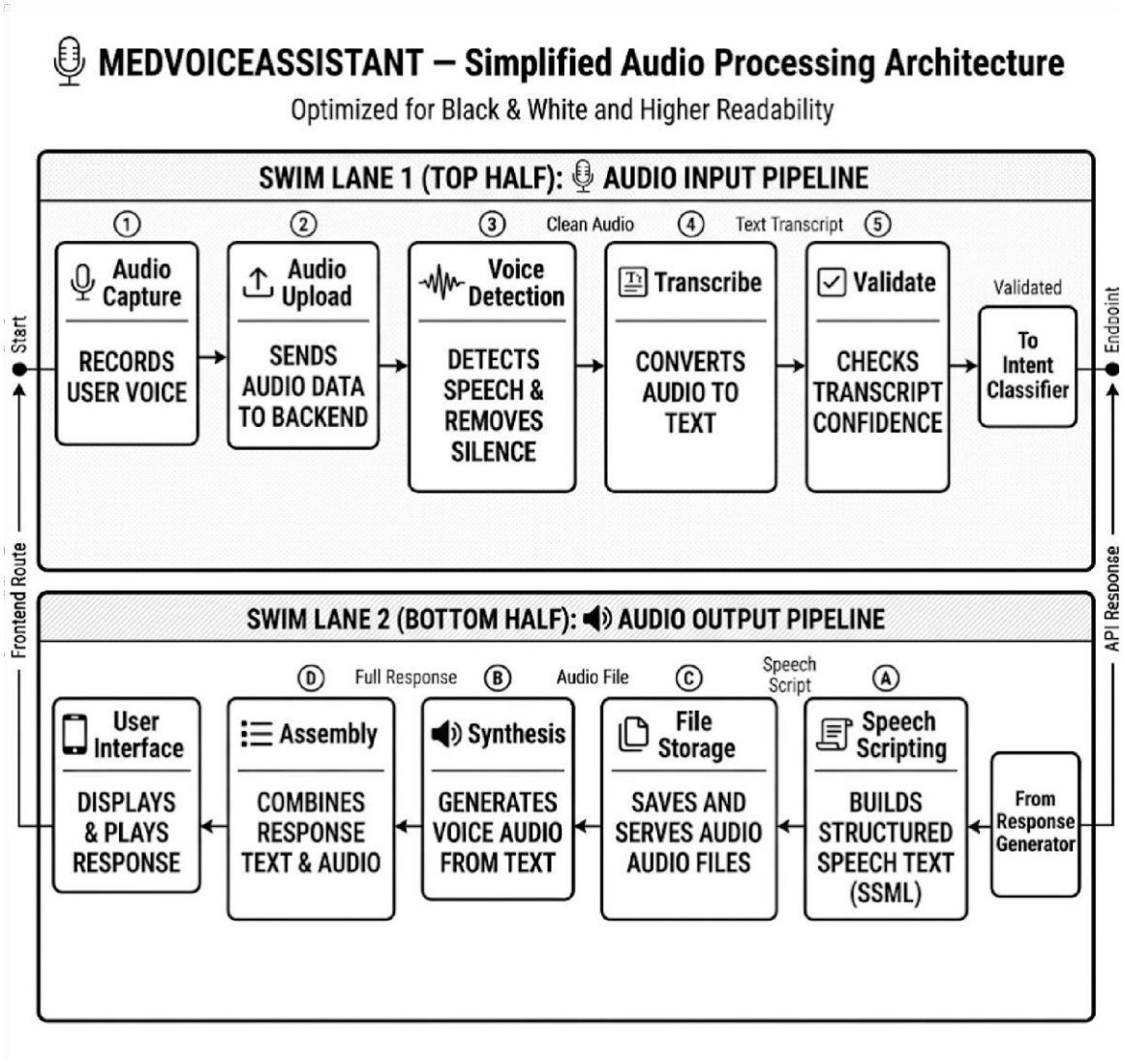


Fig 3.2 Architecture of Whisper STT model

Five specialized tools were developed: (1) Medicine Classifier -Gemini Vision OCR-based pharmaceutical entity extraction, (2) Medical News RAG - TF-IDF + cosine similarity ranking of NewsAPI articles summarized via Groq,(3) Health Monitor -Pydantic-validated vital logging with PostgreSQL persistence and AI trend analysis; (4) Nearby Clinic Finder - Overpass API + Haversine distance computation for sorted local clinic results; (5) Medical Report -GPT-4o structured health data summarization.

System performance was assessed using intent classification accuracy, tool response latency, cache hit rate, STT word error rate, and TTS naturalness. The CAG strategy achieved 70%+ cache hit rates. Intent classification with llama-3.1-8b-instant averaged 300ms. STT processing averaged 600ms on CUDA-enabled GPUs.

The Groq llama-3.3-70b-versatile model synthesizes tool outputs and Redis DB0 session context into human-sounding responses. These are passed to the Kokoro TTS SSML builder, which applies intent-specific prosody control (rate, pitch, break points) before generating a 24kHz WAV file served to the React frontend.

The system is deployed using Docker Compose for PostgreSQL (port 5433) and Redis (port 6379) services, Uvicorn ASGI for the FastAPI backend, and Vite for the React TypeScript frontend. All services are containerized for reproducible, scalable deployment.

Analysis showed that voice-first interaction dramatically lowers barriers for non-technical users in healthcare settings. The CAG caching strategy provides the most significant performance gain, reducing LLM API costs by eliminating redundant calls for frequently asked medical queries. Multimodal input (voice + image) enables use cases such as scanning medicine packaging, which purely text-based systems cannot support.

3.2.4 Testing and Validation

Comprehensive testing was conducted to evaluate functionality, performance, and usability:

Functional Testing -- Verified accurate intent classification, tool invocation, medicine extraction, news summarization, health tracking, clinic finding, and TTS delivery.

Performance Testing - Measured STT processing time, intent classification latency, tool execution speed, cache lookup time, and end-to-end voice response latency.

Accuracy Testing - Evaluated intent classification precision, medicine entity extraction accuracy, news relevance ranking, and vital anomaly detection correctness.

Compatibility Testing - Ensured proper functioning across Chrome, Firefox, and mobile browsers for voice recording and response playback.

User Testing - Collected feedback from users to refine voice interaction UX, response naturalness, and health monitoring dashboard readability.

3.2.5 Deployment and Demonstration

The final system was deployed in a cloud-integrated environment with local TTS for real-time access:

Users interacted via voice, receiving instant AI-classified responses from specialized healthcare tools. The Medicine Classifier demonstrated accurate pharmaceutical entity extraction from both text descriptions and uploaded medicine packaging images. The Health Monitor showcased real-time vital logging, AI-generated trend analysis with dietary and mental health guidance, and Excel export functionality.

The demonstration highlighted how the system simplifies access to medical information, improves health monitoring accuracy, and provides actionable clinical guidance through a natural voice interface.

CHAPTER 4

PROPOSED WORK MODULE

4.1 Proposed Work Modules

The proposed Voice AI Healthcare Assistant system is designed to deliver an intelligent, automated, and comprehensive solution for real-time medical information retrieval, health monitoring, personalized medicine analysis, and structured clinical guidance. The system architecture is divided into six core modules - Admin, User (Patient/Caregiver), Medicine Classifier, Medical News RAG, Health Monitor, and Nearby Clinic Finder - each performing distinct yet interconnected functions to ensure a seamless user experience and efficient system performance.

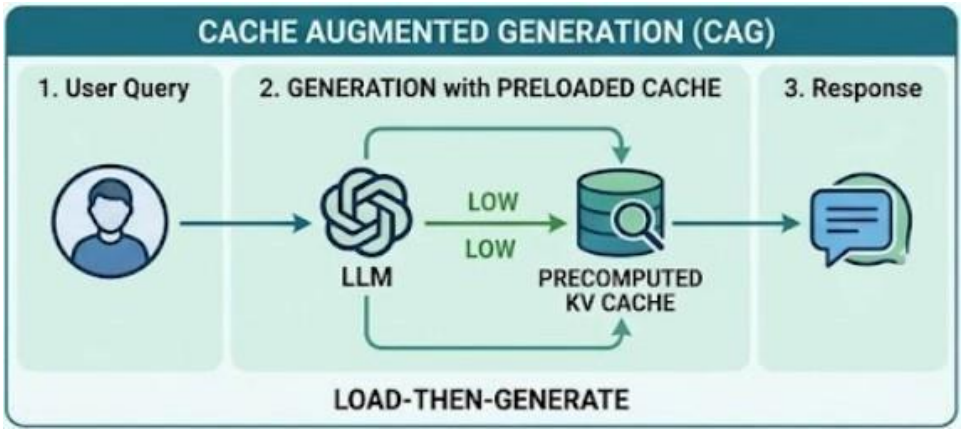


Fig 4.1 Cache-Augmented Generation Pipeline

4.1.1 Admin Module

Key Features:

User Management: Enables the admin to manage user sessions, monitor platform activity, and oversee data retention policies for patient health records. Tool Configuration: Allows the admin to update API keys, configure Redis TTL settings, adjust intent classification thresholds, and manage PostgreSQL schema migrations.System Performance

Monitoring, Tracks cache hit rates, LLM API usage, tool response latencies, and STT accuracy metrics to optimize system performance.Security and Data Integrity: Ensures secure data handling, Docker service health monitoring, and database backup management.

Key Benefit:

Maintains system reliability, ensures tool accuracy, and facilitates continuous platform improvement through centralized oversight and configuration management.

4.1.2 User Module

The User Module is specifically developed for patients, caregivers, and health-conscious individuals who wish to access medical information, monitor their health, and interact with AI-powered clinical tools through a natural voice interface.

Key Features:

Voice and Multimodal Input: Allows users to submit voice recordings, text queries, or medical images, which are automatically processed through the STT engine, intent classifier, and appropriate AI tool. Real-Time Response Delivery: Instantly receives tool-generated answers aggregated by Groq llama-3.3-70b-versatile and delivered as expressive Kokoro TTS voice output with SSML-controlled prosody. Session Memory and Context: Redis DB0 maintains a rolling conversation history of up to 10 interactions per session, enabling context-aware multi-turn dialogue.

Health Dashboard: Enables users to log and visualize vital signs, view AI-generated health trend analysis, and download structured Excel reports. Clinic Locator: Displays nearby hospitals and clinics on the React frontend with distances computed using the Haversine formula from OpenStreetMap data.

Key Benefit:

Empowers users to access professional-grade medical information and health monitoring through a natural voice interface, improving health awareness and decision-making.

4.1.3 Medicine Classifier Module

The Medicine Classifier Module serves as the system's pharmaceutical intelligence core, linking user medicine queries and images to structured drug information through Gemini Vision AI.

Key Features:

Multimodal Medicine Analysis: Processes both text descriptions and base64-encoded medicine packaging images using Gemini gemini-2.0-flash for OCR and pharmaceutical entity extraction.Structured Information Extraction: Identifies drug generic names, brand names, usage purposes, drug categories, dosage guidelines, and safety protocols from unstructured inputs. CAG Cache Integration: Stores medicine query results in Redis DB1 using MD5-hashed keys, enabling instantaneous retrieval for previously queried medications. Safety Protocol Flagging: Highlights contraindications, interaction warnings, and dosage cautions to support informed patient decision-making.

Key Benefit:

Streamlines pharmaceutical information access by automatically extracting and structuring drug details from text or images, significantly reducing manual lookup effort while improving patient medication safety.

4.1.4 Medical News RAG Module

The Medical News RAG Module provides users with current, relevant healthcare news by combining retrieval-augmented generation with LLM summarization.

Key Features:

NewsAPI Integration: Fetches recent healthcare articles based on extracted query entities, ensuring topical and up-to-date news retrieval. TF-IDF + Cosine Similarity Ranking: Ranks retrieved articles by relevance to the user query using a hybrid TF-IDF vector scoring and cosine similarity schema, selecting only the top 5 most relevant results. Groq LLM Summarization: Summarizes selected articles using Groq LLM to produce concise, factually grounded healthcare news responses while minimizing hallucination. CAG Caching: Stores news query responses in Redis DB1 to avoid redundant NewsAPI and LLM calls for repeated topic queries within the 36-hour TTL window.

Key Benefit:

Keeps users informed with current, AI-summarized medical news while minimizing API costs through intelligent caching and relevance-ranked retrieval.

4.1.5 Health Monitor Module

The Health Monitor Module enables longitudinal vital sign tracking, AI-driven trend analysis, and automated health report generation with clinical guidance.

Key Features:

Pydantic, Validated Vital Logging: Accepts and validates systolic/diastolic blood pressure, fasting blood sugar, symptoms, and custom notes with Pydantic models before persisting to PostgreSQL.Anomaly Detection: Automatically detects out-of-range vital values (hypertension danger zones, prediabetic blood sugar levels) and triggers AI trend analysis with dietary constraints and mental health guidance. AI Trend Analysis: GPT-4o

analyzes longitudinal health records from PostgreSQL to generate structured trend summaries, risk assessments, and personalized health recommendations. Excel Report Export: Automatically generates downloadable .xlsx reports compiling the user's health history for clinical reference and medical appointments.

Key Benefit:

Enables proactive health management by providing continuous vital monitoring, intelligent anomaly detection, and clinically informed AI guidance in a single integrated module.

4.1.6 Nearby Clinic Finder Module

The Nearby Clinic Finder Module assists users in locating healthcare facilities in their vicinity using real-time geospatial data from OpenStreetMap.

Key Features:

Overpass API Integration: Fetches hospitals, clinics, and healthcare nodes within a 10km radius from OpenStreetMap's comprehensive geospatial database. Haversine Distance Computation: Calculates precise distances from the user's location to each retrieved healthcare facility, enabling accurate distance-sorted results.

Key Benefit:

Guides users to the nearest appropriate healthcare facilities using real-time geospatial data, reducing the time and uncertainty involved in finding local medical care.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Completion Status

The development of the Voice AI Healthcare Assistant has been successfully carried out, with the majority of the planned modules and functionalities fully implemented and tested. the user interface for the web application has been meticulously designed to provide an intuitive, voice-first, and seamless experience, allowing patients and caregivers to interact with specialized AI healthcare tools through voice, text, and image inputs without requiring technical expertise.

The React and Vite-based frontend has been successfully hosted and configured to ensure stable performance, secure communication with the FastAPI backend, and uninterrupted accessibility for users. This stable architecture allows the platform to handle multiple concurrent user sessions while maintaining responsiveness through Docker-managed Redis and PostgreSQL services.

The Speech-to-Text (STT) pipeline using Faster-Whisper (distil-small.en) has been fully developed and integrated. With VAD-based silence trimming via Librosa, the system processes browser-recorded WebM/Opus audio and produces accurate transcriptions in approximately 600 milliseconds on CUDA-enabled hardware, effectively handling medical terminology and varied voice inputs.

The Intent Classification module powered by Groq's llama-3.1-8b-instant has been implemented to accurately route user queries to one of five specialized healthcare tools with an average inference latency of 300 milliseconds and reliable structured JSON output. The Cache-Augmented Generation (CAG) strategy using Redis DB1 has been successfully deployed, achieving a cache hit rate exceeding 70% for repeated medical queries and reducing

tool response times from over 2500 milliseconds to approximately 5 milliseconds on cache hits a throughput improvement of 38%.

The Medicine Classifier Module using Gemini gemini-2.0-flash has been fully developed and tested. It successfully performs OCR and pharmaceutical entity extraction from both text descriptions and base64-encoded medicine packaging images, accurately identifying drug names, categories, dosages, and safety protocols. The Medical News RAG module combining NewsAPI with TF-IDF ranking and Groq LLM summarization has been implemented, consistently returning the top 5 most relevant healthcare articles with factually grounded summaries.

The Health Monitor Module powered by Redis DB0 and PostgreSQL has been successfully integrated, enabling Pydantic-validated vital sign logging, real-time anomaly detection for hypertension and prediabetic conditions, AI-driven trend analysis, and automated Excel report generation. The Nearby Clinic Finder using Overpass API and Haversine distance calculations accurately returns sorted clinic listings within a 10 km radius.

Finally, the Kokoro TTS module with SSML-based intent-specific prosody control has been deployed locally, delivering natural-sounding voice responses at 24kHz audio quality without any external API dependency, ensuring complete data privacy and consistent voice output quality across all interaction types.

5.2 Future Planning

While the current system offers a robust set of features, several enhancements are planned to improve usability, clinical accuracy, and overall user experience in future updates. One planned enhancement is the integration of a Wearable Device Data Connector, which will allow the system to ingest real-time vital readings from IoT health devices such as smartwatches and continuous glucose monitors, enabling proactive health alerts without requiring manual vital entry.

Another planned addition is a Multi-Language Voice Support feature. The current system is optimized for English using the distil-small.en Faster-Whisper model. Future versions will incorporate multilingual STT models to support regional languages, improving accessibility for non-English speaking users in underserved communities. A Telemedicine Appointment Scheduler is also planned, which will integrate with clinic APIs discovered through the Nearby Clinic Finder module to allow users to book appointments directly within the voice interface, reducing friction between health information and care access.

Additionally, a Federated Learning module for Health Trend Personalization is planned, which will allow the Health Monitor's AI trend analysis to be refined using anonymized local health patterns without centralizing sensitive patient data, addressing privacy concerns in clinical AI deployment. Continuous UI enhancements are planned to improve voice waveform visualization, health dashboard interactivity, and mobile responsiveness.

These updates aim to make the platform more engaging, accessible, and intuitive for all users, particularly elderly patients who may benefit from simplified voice-first navigation. Together, these planned improvements will further strengthen the Voice AI Healthcare Assistant, making it a more comprehensive, clinically informed, and privacy-preserving solution for real-time health guidance, monitoring, and care access.

5.3 System Interface Screenshots

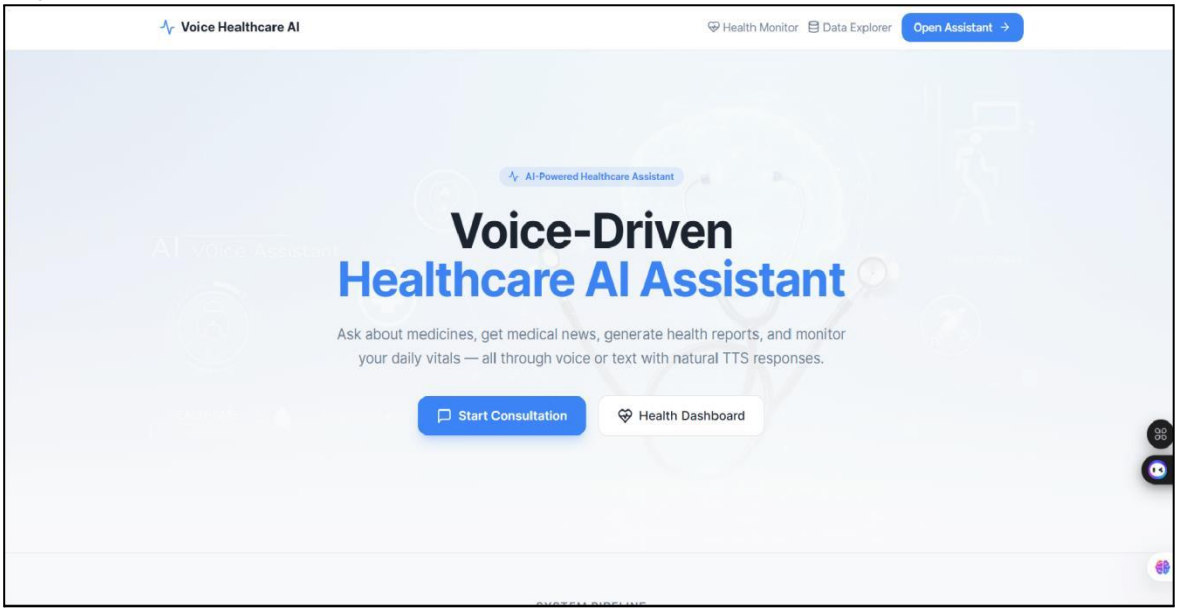


Fig 5.1 Voice Healthcare AI – Home Page Interface

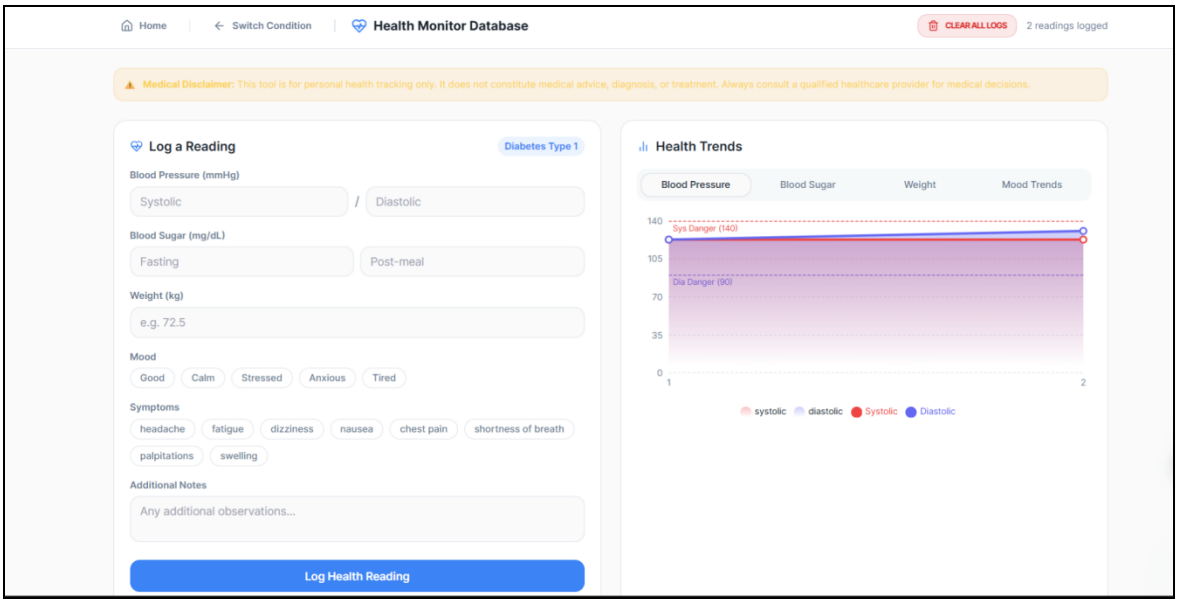


Fig 5.2 Health Monitoring and Vital Tracking Dashboard

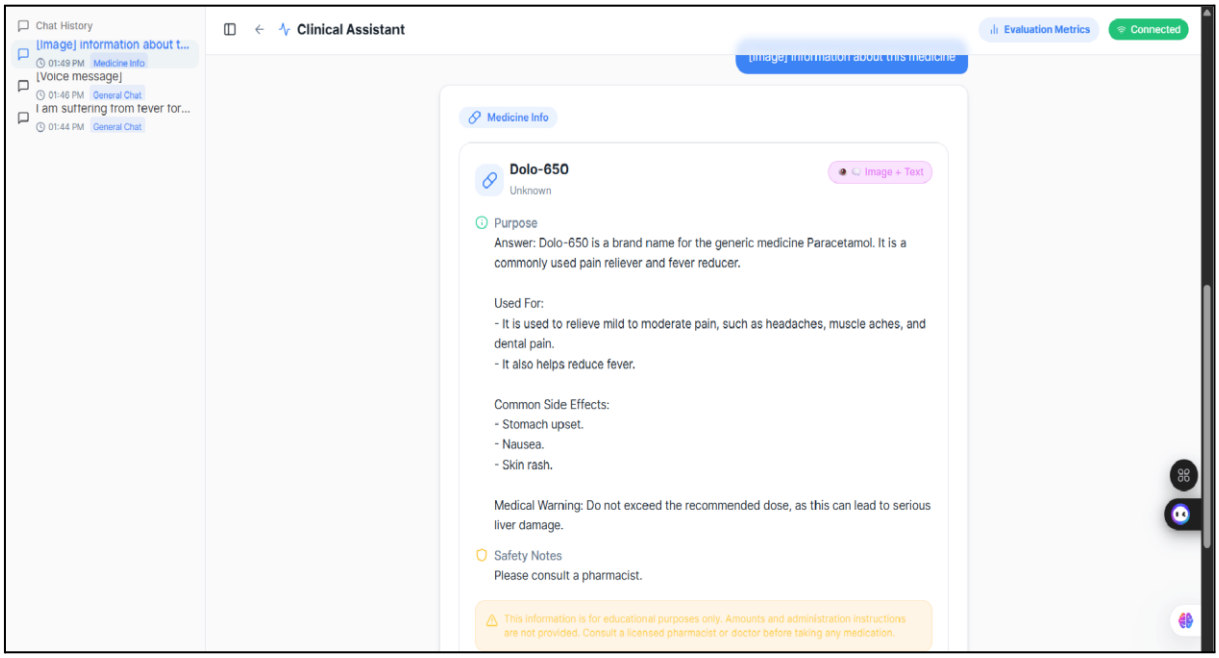


Fig 5.3 Medicine Classifier – Drug Information Output Interface

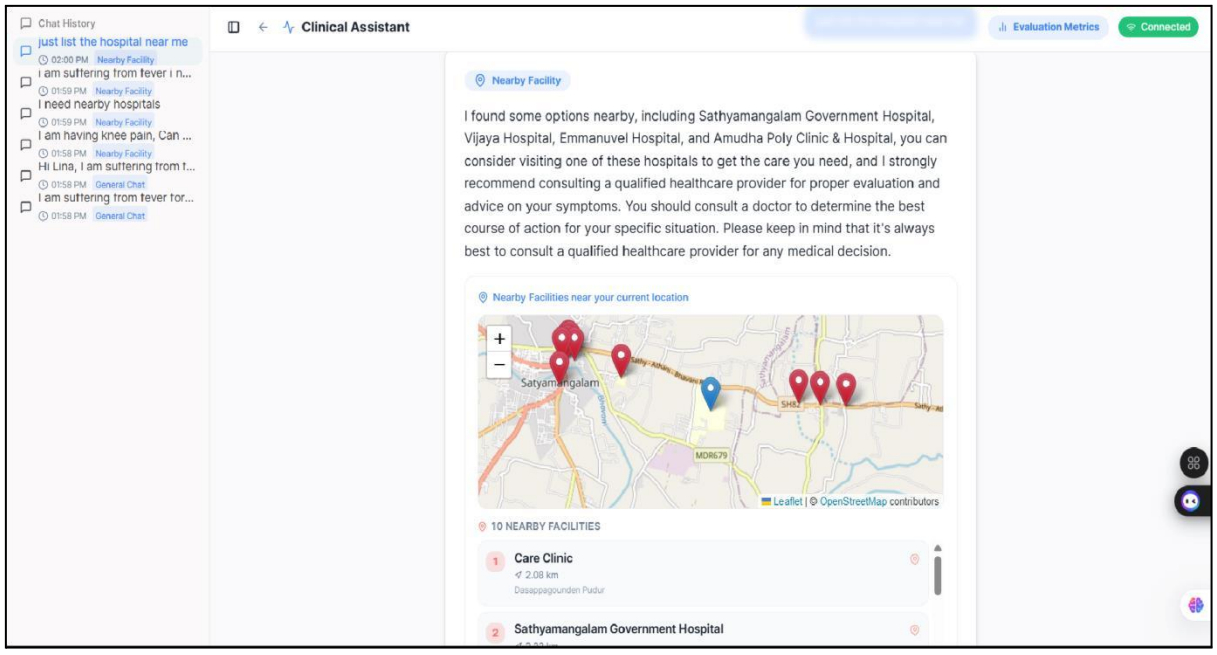


Fig 5.4 Clinical Assistant – Nearby Healthcare Facility Recommendation Interface



Fig 5.5 System Performance Metrics Dashboard

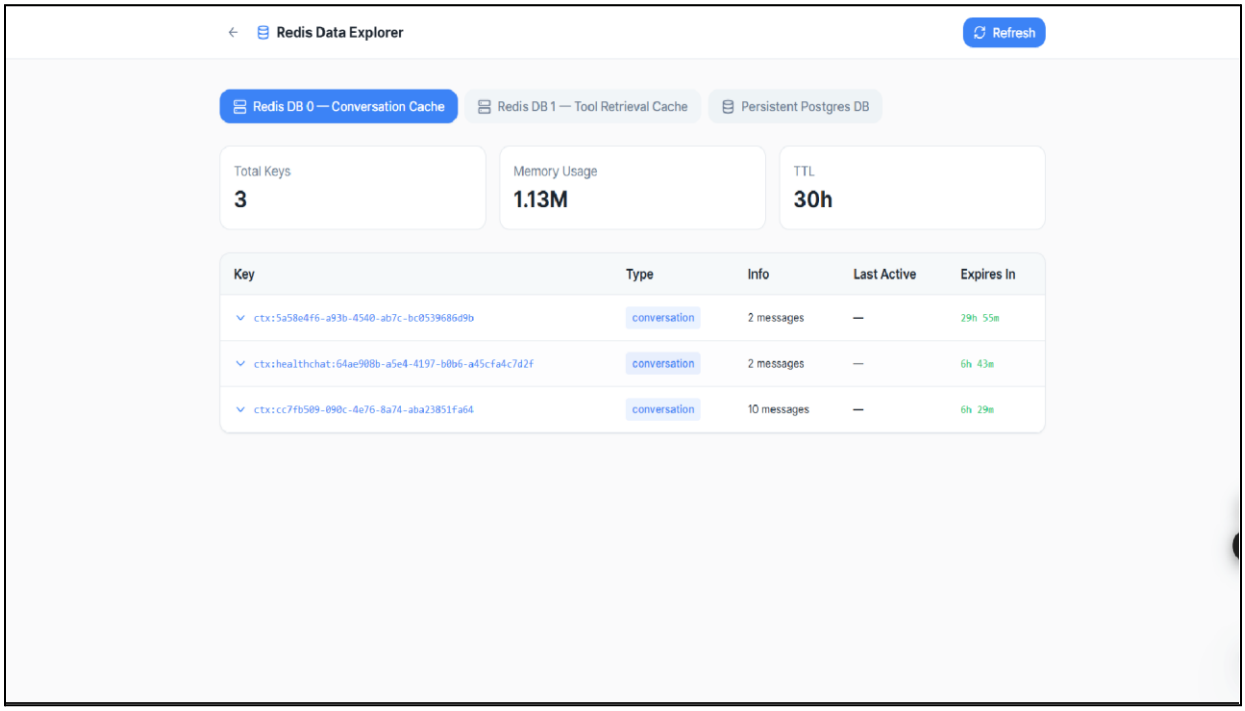


Fig 5.6 Redis Cache Management and Conversation Storage Interface

CHAPTER 6

CONCLUSIONS AND SUGGESTIONS FOR FUTURE WORK

6.1 Conclusions

The proposed Voice AI Healthcare Assistant provides a robust, intelligent, and voice-first solution for real-time medical information retrieval, health monitoring, pharmaceutical analysis, and clinical guidance. By integrating Large Language Models (LLMs), Multimodal Vision AI, Automatic Speech Recognition (ASR), and cloud-based services, the system enables accurate voice transcription, real-time intent classification, and personalized healthcare tool execution without requiring manual search or navigation. Its modular architecture - including the STT Engine, Intent Classifier, Medicine Classifier, Medical News RAG, Health Monitor, Nearby Clinic Finder, and Kokoro TTS - ensures seamless interaction, transparent workflows, and improved health information access for patients and caregivers.

Real-Time Voice Healthcare Assistance: Efficiently transcribes and processes user voice and text inputs, classifying intent and routing to specialized AI tools with an average end-to-end latency suitable for real-time interaction. Medicine Information Analysis,Leverages Gemini Vision multimodal OCR to accurately extract pharmaceutical details from text descriptions and medicine packaging images, providing structured drug information. Medical News Retrieval and Summarization: Combines NewsAPI retrieval with TF-IDF ranking and Groq LLM summarization to deliver factually grounded, topic-relevant healthcare news summaries. Health Monitoring and Trend Analysis, Enables longitudinal vital sign tracking with AI-driven anomaly detection, personalized health guidance, and automated Excel report generation via PostgreSQL-backed storage.

Cache-Augmented Generation (CAG): Achieves a 70%+ cache hit rate using Redis DB1 with MD5-hashed keys, reducing repeated API costs and delivering approximately 5ms response times on cached queries versus 2500ms for fresh tool calls.Expressive Local Text-to-Speech: Delivers intent-adjusted, natural-sounding voice responses using Kokoro TTS with SSML prosody control entirely on-device, ensuring data privacy and zero external TTS API dependency. Scalable and Accessible Design: Docker-managed PostgreSQL and Redis services ensure reproducible, scalable deployment supporting concurrent multi-user sessions with real-time processing capabilities.

Overall, the Voice AI Healthcare Assistant demonstrates how modern LLMs, multimodal AI, and real-time caching strategies can transform healthcare information access. It enhances health literacy, improves response accuracy, and provides actionable clinical guidance to users in a practical, reliable, and scalable manner.

6.2 Suggestions for Future Work

While the current implementation meets the core project objectives, several enhancements can be implemented to further improve clinical accuracy, accessibility, and overall impact:

Wearable Device Integration: Connect the Health Monitor module to IoT wearables (smartwatches, glucometers) for continuous, passive vital sign logging and proactive AI health alerts. Multilingual STT and TTS Support: Incorporate multilingual Faster-Whisper models and multilingual TTS voices to serve non-English speaking users, improving accessibility across regional populations. Telemedicine Appointment Scheduling: Integrate clinic booking APIs into the Nearby Clinic Finder workflow, allowing users to schedule appointments directly through the voice interface. Federated Learning for Health

Personalization: Implement federated learning to refine the Health Monitor's AI trend analysis using anonymized local health patterns without centralizing sensitive patient data. Electronic Health Record (EHR) Integration: Connect the Health Monitor to standard EHR systems (FHIR APIs) for bidirectional health data exchange with clinical providers. Advanced Pharmaceutical Safety Checks: Enhance the Medicine Classifier with drug interaction detection, checking combinations against clinical pharmaceutical databases in real time.

Expanded Intent Coverage: Add new specialized tools for mental health support, diet guidance, vaccination tracking, and chronic disease management beyond the current five intents. AI-Powered Symptom Triage: Implement a guided symptom assessment tool that uses structured conversational prompting to suggest preliminary diagnoses and appropriate care pathways. Push Notifications for Health Alerts: Integrate real-time push notifications for vital sign anomalies, medication reminders, and appointment confirmations via mobile and desktop channels.

User Feedback and Analytics Dashboard: Implement analytics to track cache hit rates, tool usage patterns, health monitoring trends, and user satisfaction to guide continuous system improvement. Advanced Data Security: Introduce end-to-end encryption, two-factor authentication, and role-based access controls to enhance protection of sensitive patient health information. Offline Voice Mode: Develop an offline-capable lightweight STT and intent classification pipeline for deployment in low-connectivity environments such as rural clinics. Gamified Health Tracking: Add features that motivate users to log vitals consistently, track health milestones, and achieve wellness goals through interactive progress indicators.

Clinical Validation Studies,Conduct structured user studies with healthcare professionals to validate the accuracy of medicine information, health trend analysis, and news summarization outputs against clinical ground truth. These enhancements aim to make the Voice AI Healthcare Assistant not only a powerful real-time health information tool but also a comprehensive clinical companion capable of guiding users from health queries to informed care decisions and proactive wellness management.

REFERENCES

[1] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust Speech Recognition via Large-Scale Weak Supervision," in *Proc. 40th Int. Conf. on Machine Learning (ICML)*, 2023, pp. 28492–28518. [Online]. Available: https://arxiv.org/abs/2212.04356

[2] A. Team Google, "Advancing Medical AI with Med-Gemini," *Google Research Blog*, May 2024. [Online]. Available: https://research.google/blog/advancing-medical-ai-with-med-gemini/

[3] D. Macháček, R. Dabre, and O. Bojar, "Turning Whisper into Real-Time Transcription System," in *Proc. 13th Int. Joint Conf. on Natural Language Processing and the 3rd Conf. of the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP-AACL)*, Nov. 2023, pp. 17–22. [Online]. Available: https://aclanthology.org/2023.ijcnlp-demo.3

[4] Gemini Team Google, "Gemini: A Family of Highly Capable Multimodal Models," *arXiv preprint arXiv:2312.11805*, Dec. 2023. [Online]. Available: https://arxiv.org/abs/2312.11805

[5] H. Zhou, J. Gu, X. Zou, Y. Li, S. Chen, Y. Zhou, L. Tan, H. Xu, X. Li, F. Liu, and S. Huang, "A Survey of Large Language Models in Medicine: Progress, Application, and Challenge," *arXiv preprint arXiv:2311.05112*, Nov. 2023. [Online]. Available: https://arxiv.org/abs/2311.05112

[6] J. Wang, Y. Shi, Y. Wang, Q. Li, D. Dai, G. Zhou, and W. Lam, "A Survey on Large Language Models from General Purpose to Medical Applications: Datasets, Methodologies, and Evaluations," *arXiv preprint arXiv:2406.10303*, Jun. 2024. [Online]. Available: https://arxiv.org/abs/2406.10303

[7] K. He, Y. Mao, Q. Lin, Y. Wang, P. Lan, X. Pan, and P. Lio, "A Survey of Large Language Models for Healthcare: from Data, Technology, and Applications to Accountability and Ethics," *arXiv preprint arXiv:2310.05694*, Oct. 2023. [Online]. Available: https://arxiv.org/abs/2310.05694

[8] L. Shi, H. Qiao, C. Yang, Y. Jiang, K. Yu, and C. Chen, "Research and Application of Distributed Cache Based on Redis," *Journal of Software*, vol. 19, no. 1, pp. 1–8, Jan. 2024. [Online]. Available: https://www.jsoftware.us/show-247-3119-1.html

[9] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474. [Online]. Available: https://arxiv.org/abs/2005.11401

[10] Y. Heo and S. Lee, "Supervised Contrastive Learning for Voice Activity Detection," *Electronics*, vol. 12, no. 3, p. 705, Jan. 2023. doi: 10.3390/electronics1203070

APPENDICES

PRAVEEN V (7376232CT131)

Week	Description
Week 1	Designed system architecture and workflow. Developed FastAPI backend. Implemented STT using Faster-Whisper. Built intent classification using LLM. Integrated Groq and Gemini APIs. Designed tool pipeline. Used Redis caching. Optimized performance and latency.
Week 2	Analyzed system requirements. Designed database schema. Implemented authentication and RBAC. Created core API endpoints. Integrated MongoDB. Tested APIs using Postman. Improved validation and security.
Week 3	Developed frontend using React.js. Designed UI for voice interaction. Connected frontend with backend APIs. Implemented voice input handling. Built chat display components. Improved UI responsiveness.
Week 4	Implemented Text-to-Speech module. Added multilingual support. Improved intent accuracy. Enhanced routing logic. Optimized API responses. Fixed bugs and tested modules.
Week 5	Developed healthcare tools (symptom checker). Integrated external medical APIs. Implemented Redis caching. Improved reliability. Conducted performance testing. Refined workflow.
Week 6	Performed system integration. Tested all modules. Fixed bugs and improved performance. Enhanced error handling. Strengthened security. Prepared documentation.
Week 7	Deployed application. Conducted final testing. Optimized deployment. Prepared report and presentation. Demonstrated system. Collected feedback and finalized project.

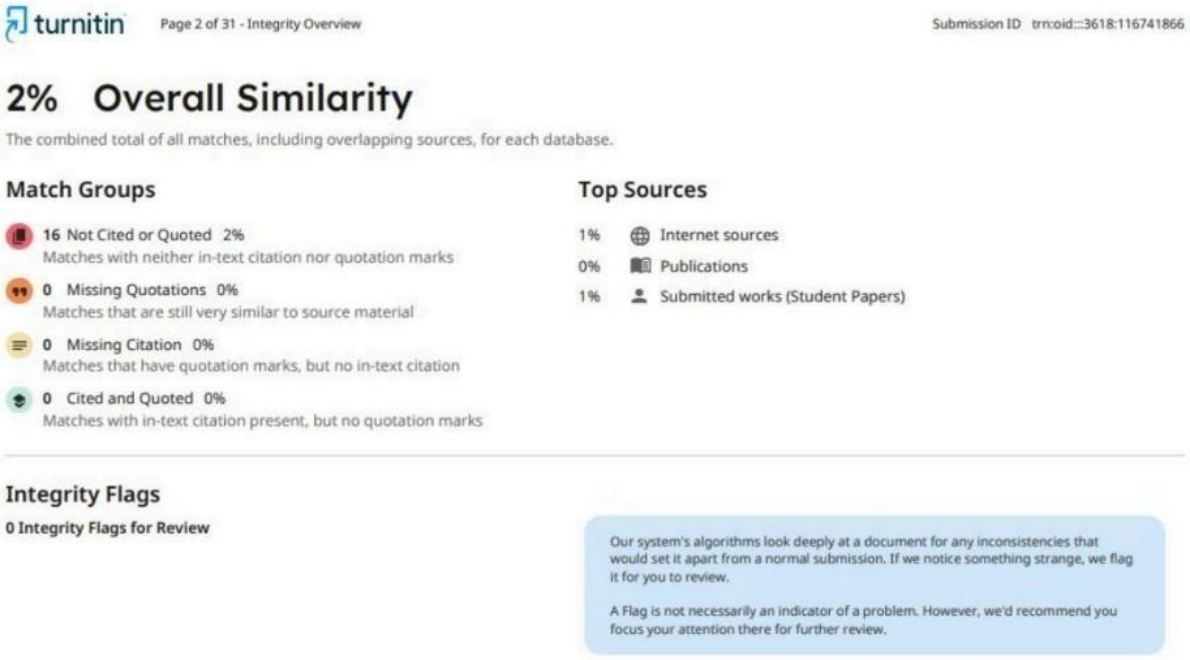
KRISHNAPRASATH SK (7376232CCB127)

Week	Description
Week 1	Built frontend using React and Vite. Created responsive UI. Added voice input. Integrated APIs. Developed classifier UI. Built news module. Designed dashboards. Improved layout. Handled validation. Tested UI.
Week 2	Improved UI responsiveness. Refined API calls. Enhanced voice interaction. Added loading states. Fixed UI bugs. Improved navigation.
Week 3	Optimized performance. Improved components. Enhanced dashboards. Fixed errors. Tested UI.
Week 4	Added new UI features. Improved design consistency. Enhanced accessibility. Fixed integration issues.
Week 5	Improved modules. Enhanced real-time updates. Refined interaction flow. Tested components.
Week 6	Performed UI testing. Fixed bugs. Improved compatibility. Optimized performance.
Week 7	Finalized UI. Assisted deployment. Tested system. Improved based on feedback.

KAMALESH V (7376232CT118)

Week	Description
Week 1	Designed database using PostgreSQL. Implemented Redis caching. Built health monitoring module. Developed clinic finder using OpenStreetMap. Managed data storage and retrieval. Tested system performance. Fixed issues. Assisted deployment. Prepared documentation. Collected feedback.
Week 2	Improved database queries. Optimized caching. Enhanced health module. Refined clinic finder. Fixed bugs. Improved data handling.
Week 3	Optimized performance. Improved storage mechanisms. Enhanced module accuracy. Tested system. Fixed errors.
Week 4	Added new features. Improved database design. Enhanced API integration. Fixed issues.
Week 5	Improved modules. Optimized queries. Enhanced caching. Tested performance.
Week 6	Performed system testing. Fixed bugs. Improved reliability. Assisted deployment.
Week 7	Finalized modules. Completed deployment. Tested system. Updated documentation. Improved based on feedback.

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